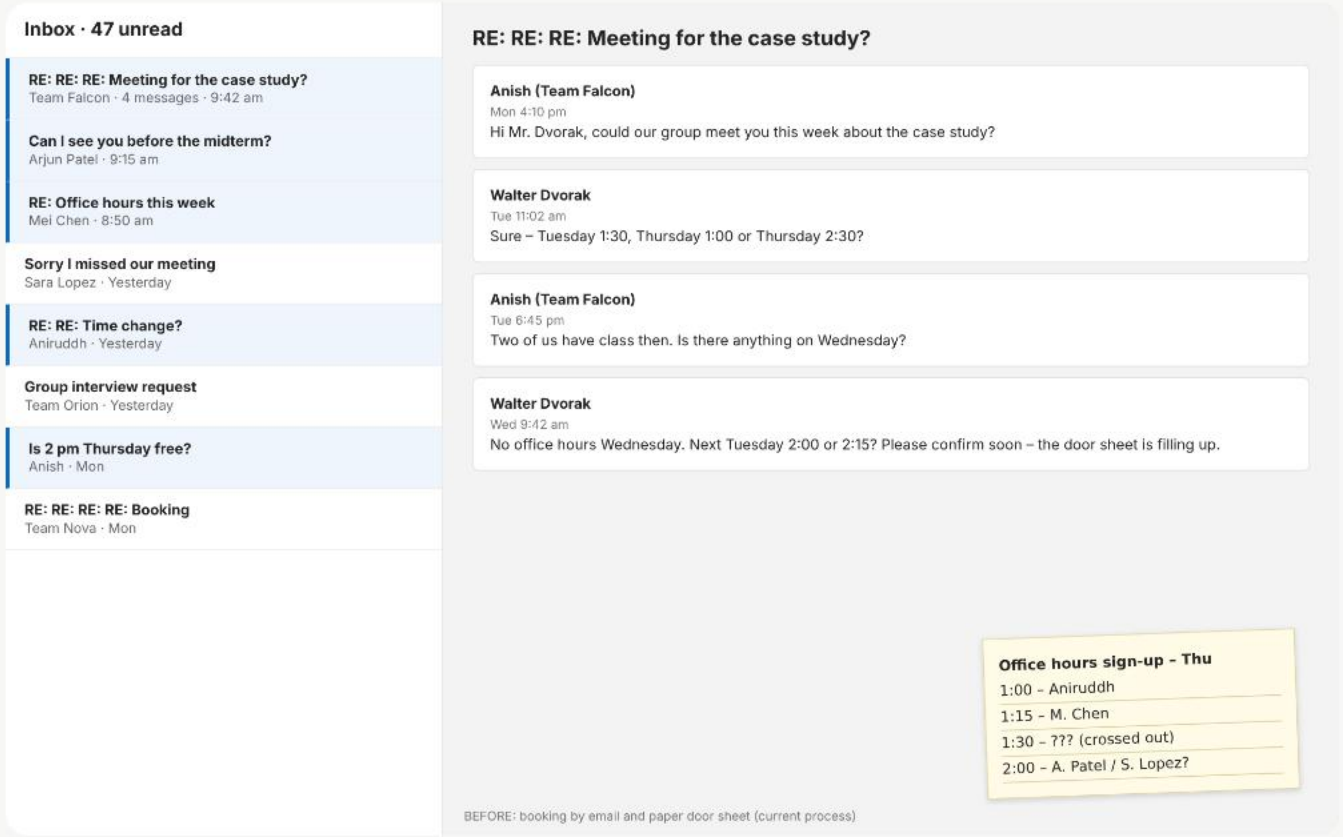
SYSTEMS ANALYSIS AND DESIGN CASE STUDY · DELIVERABLE 07

Instructor Consultation Booking System

Prototype presentation: from email back-and-forth to one-step self-booking

Prepared for Walter Dvorak, Instructor, Business Information Systems · By Md Alamgir Hossain and team

Today, booking one meeting takes days of email



BEFORE: booking by email and paper door sheet (current process)

2–4

rounds of email per booking

3–4 h

instructor time lost per week

15%

of booked meetings missed

The prototype replaces the email loop with one online step

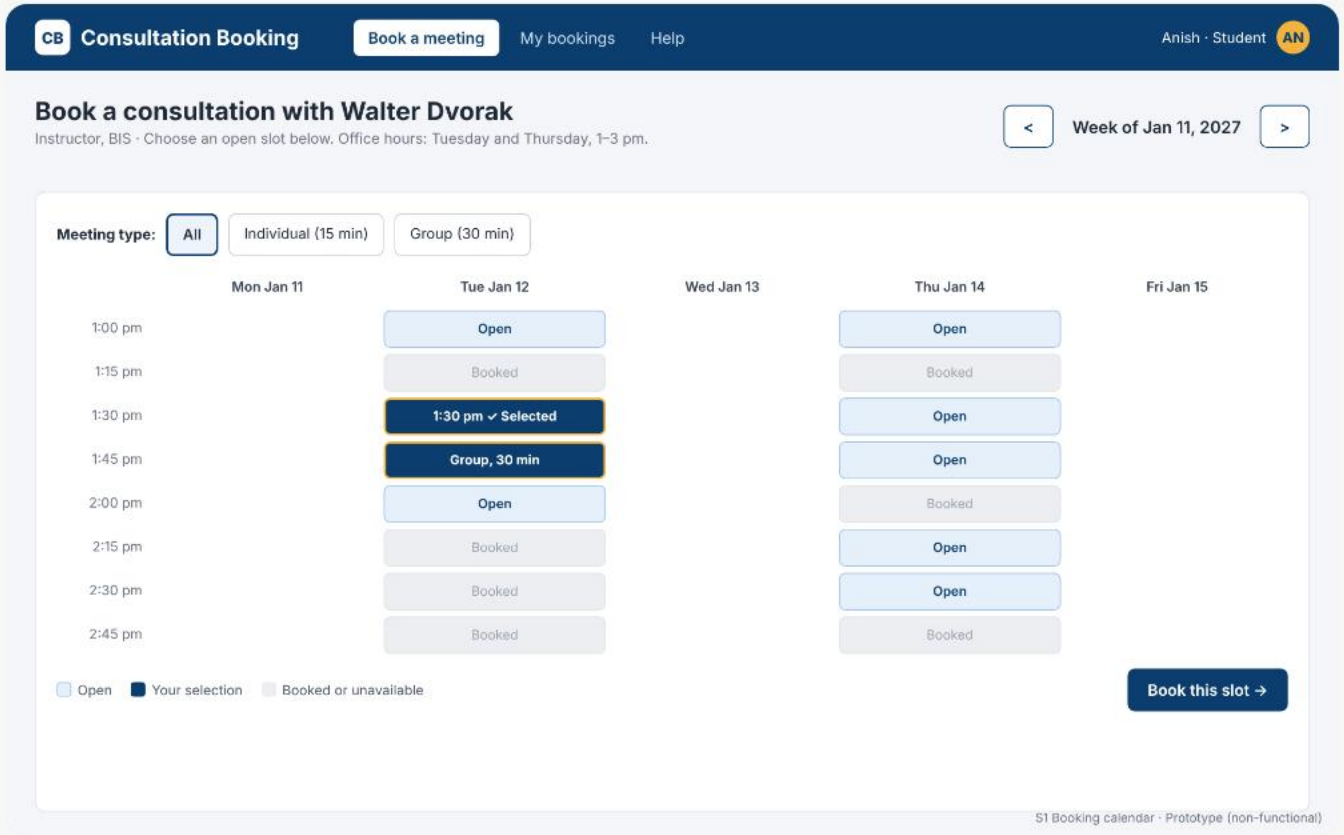
Before: email and paper

1. Student emails a request
2. Walter replies with 2–3 times
3. Back and forth until a time works
4. Walter adds it to Outlook by hand
5. No reminder; notes often skipped

After: self-booking prototype

1. Student sees open slots (S1)
2. Picks one and confirms (S2)
3. Outlook and Teams update automatically
4. Reminder sent 24 hours before
5. Walter records notes and status (S6)

S1 Booking calendar: students see every open slot at once

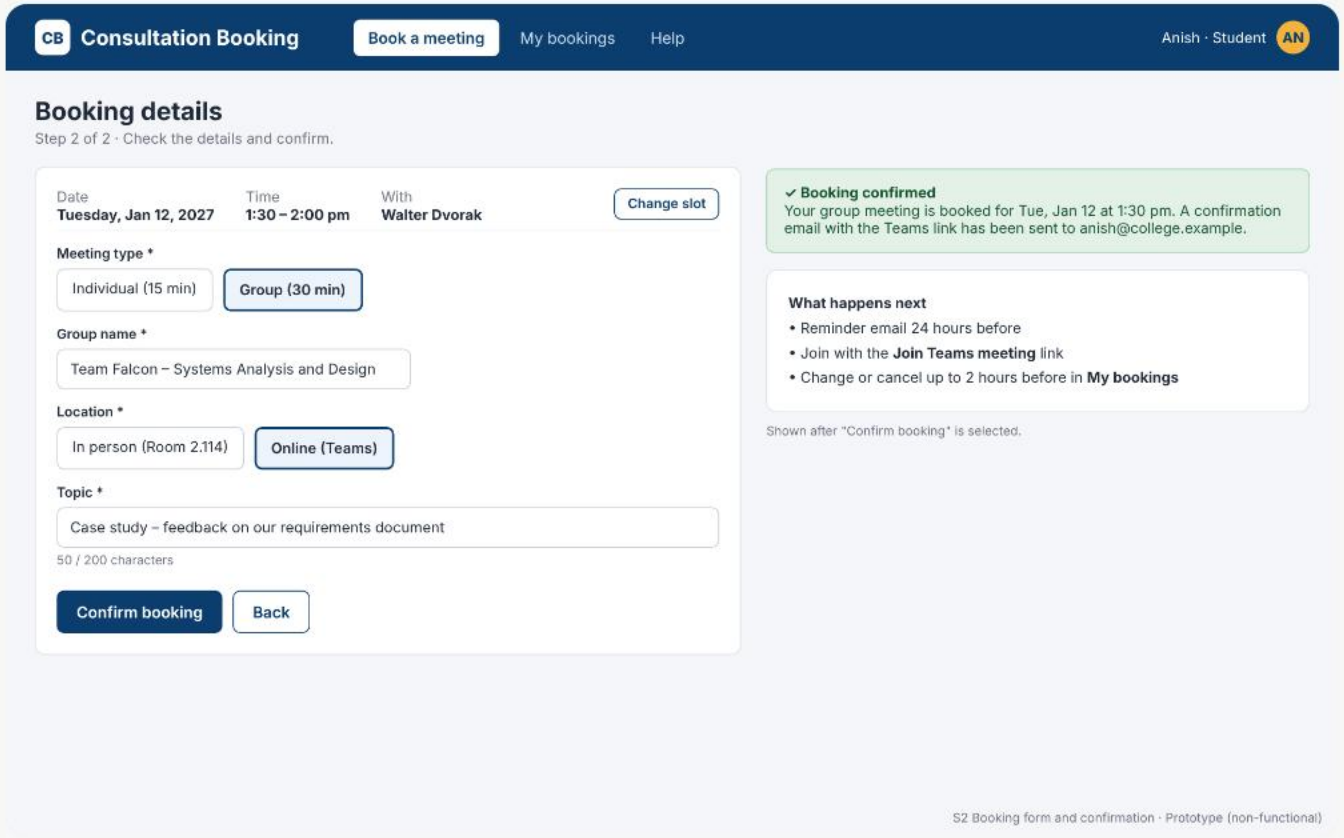


IMPROVEMENTS

- Open slots in colour; booked slots dimmed and not clickable
- Meeting-type filter: individual or group
- Week arrows to browse ahead
- Legend explains every colour

Covers FR-01, FR-04 · Ch 8: dim unavailable commands; clear labels

S2 Booking form: three choices, then instant confirmation



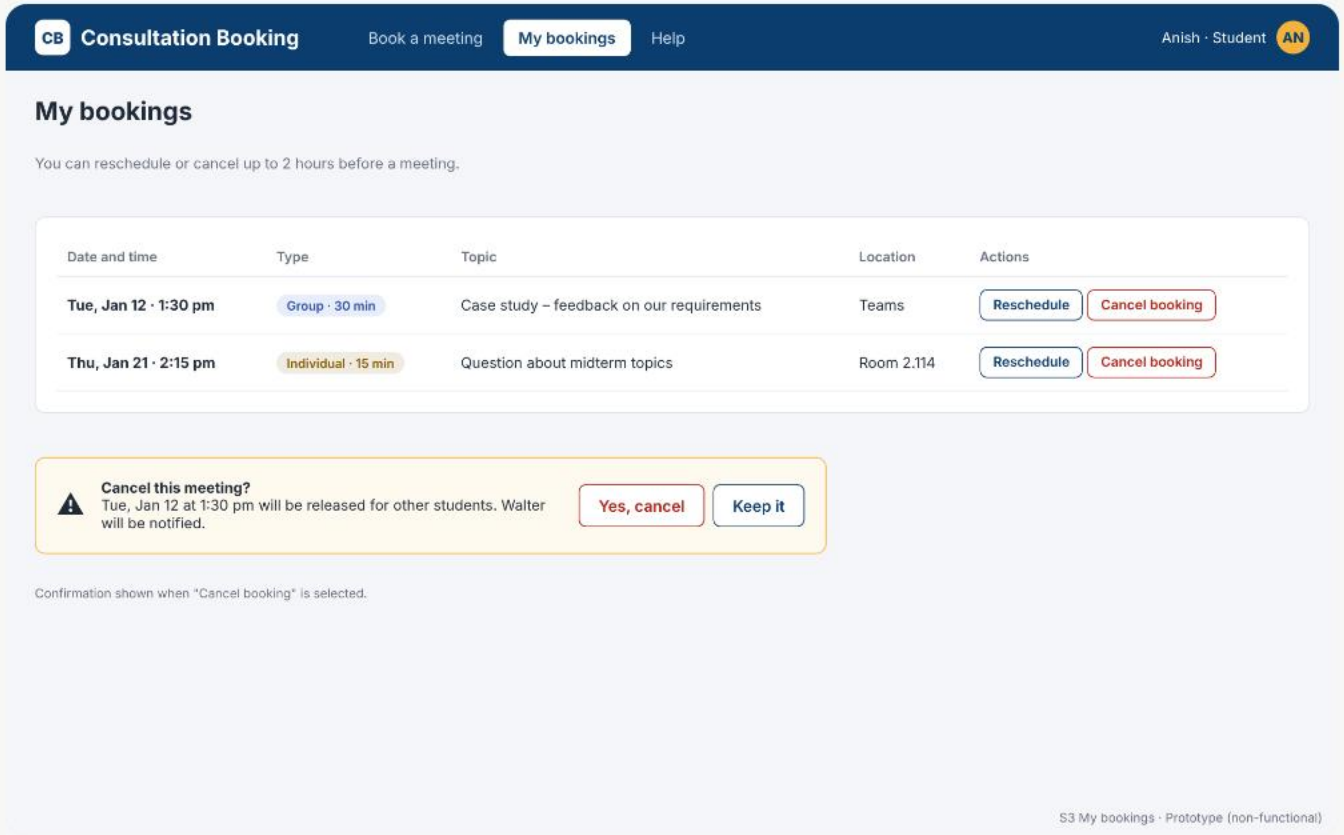
S2 Booking form and confirmation · Prototype (non-functional)

IMPROVEMENTS

- Name and email come from the college login, so nothing to retype
- Required fields marked *, with a character counter
- Teams link created automatically for online meetings
- Green confirmation explains what happens next

Covers FR-03, FR-05, FR-11 · Ch 8: minimize input, provide feedback

S3 My bookings: change or cancel without sending an email



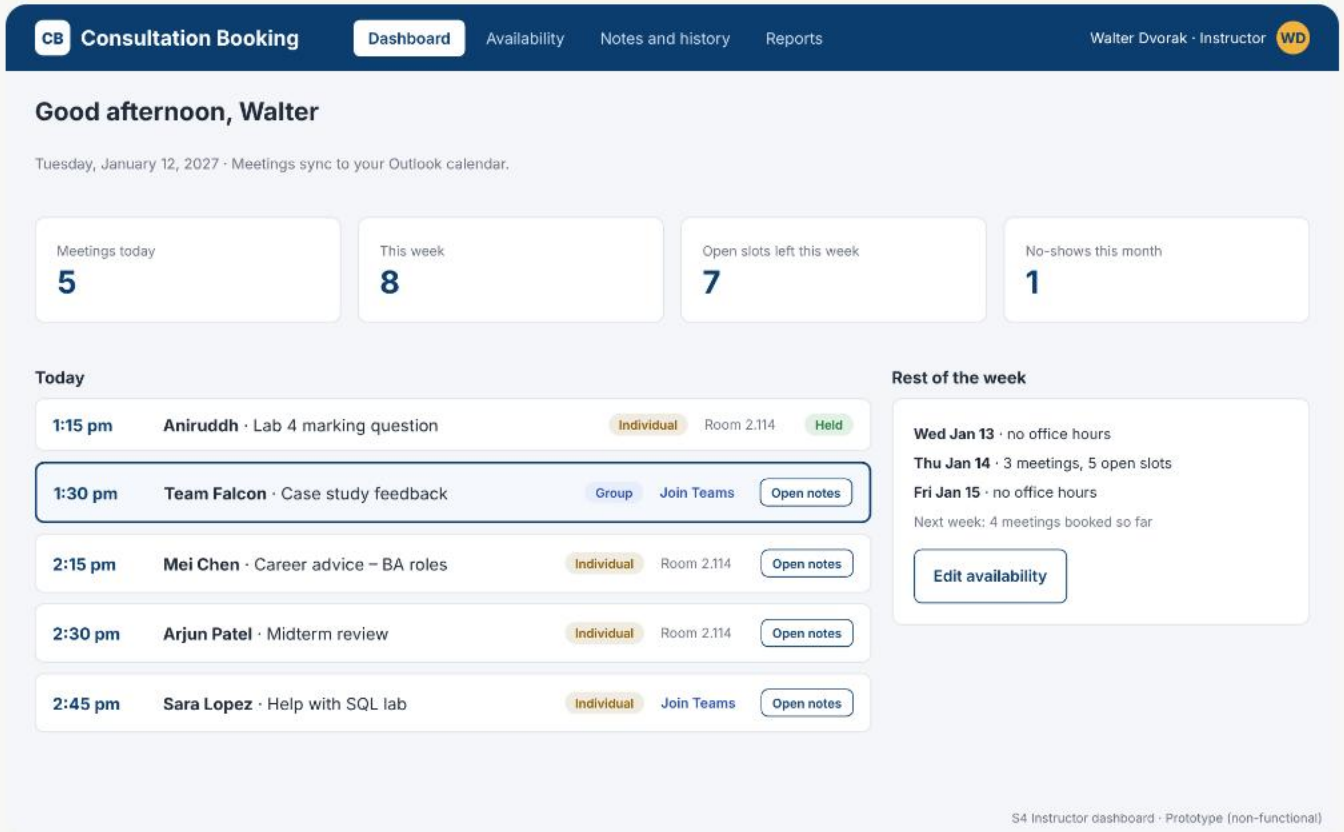
S3 My bookings · Prototype (non-functional)

IMPROVEMENTS

- Every upcoming meeting in one list
- Reschedule and Cancel on each row
- Cancel asks for confirmation first
- 2-hour cut-off stated at the top

Covers FR-06, FR-07 · Ch 8: let users correct errors easily

S4 Instructor dashboard: the whole day at a glance



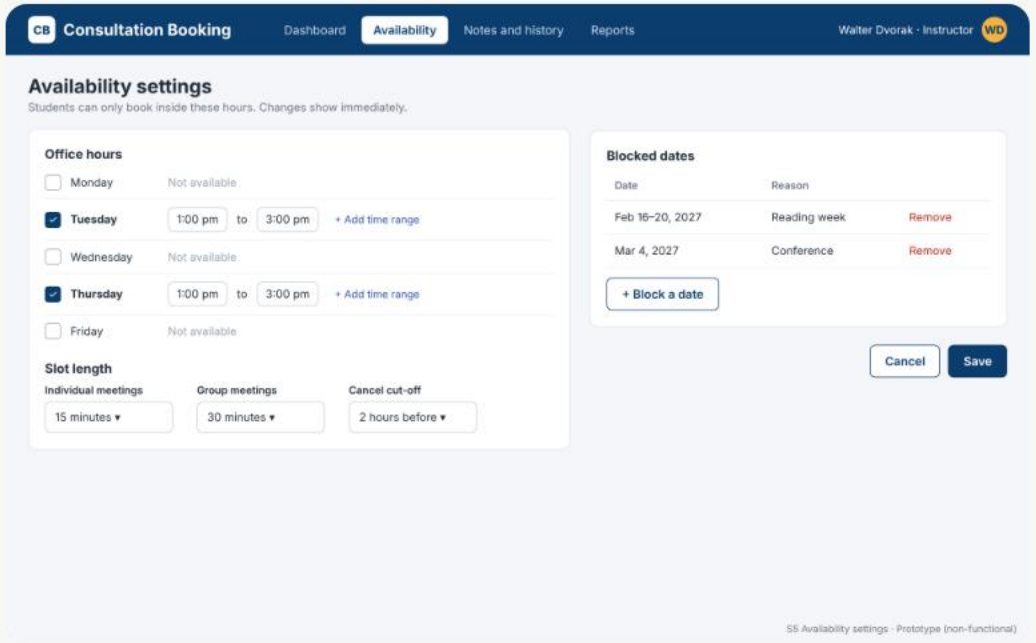
S4 Instructor dashboard · Prototype (non-functional)

IMPROVEMENTS

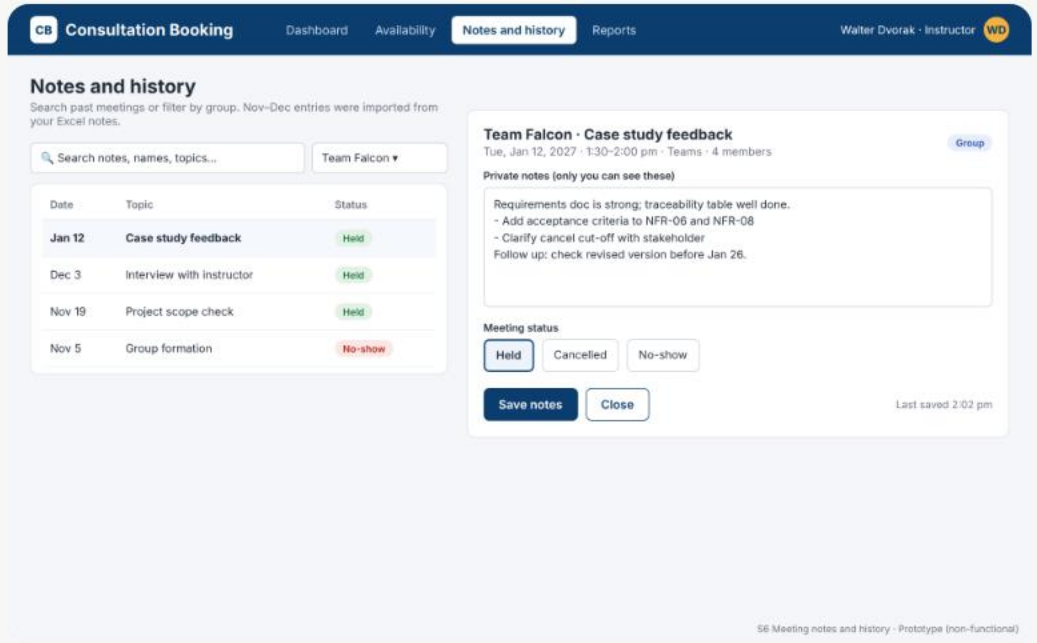
- Four summary counts across the top
- Today’s meetings in time order, with topic and location
- One-click Join Teams and Open notes
- Synced to Outlook automatically

Covers FR-08, FR-12 · Ch 8: focus on basic objectives

S5 and S6: Walter controls his hours and keeps private notes



S5 Availability: office hours, slot lengths, cancel cut-off and blocked dates in one place (FR-02)



S6 Notes and history: private notes, Held / Cancelled / No-show status, search and group filter (FR-09, FR-10, FR-13)

How we worked: the seven habits of successful interface designers

Understand the business
Interviewed Walter; CATWOE and fishbone analysis

Maximize graphics
Colour-coded slots with a legend; status pills

Think like a user
Surveyed 30 students; designed for phones too

Use models and prototypes
Use case diagram, then screens S1–S6

Focus on usability
Main option on the opening screen; few choices per screen

Invite feedback
Walter reviews and signs off; student test group

Document everything
Requirements (D05) and user manual (D06)

Prototype type
Design (throwaway) prototype: confirms requirements, then Bookings is configured

The prototype applies all 10 user interface design guidelines

Guideline (Chapter 8)	Where the prototype applies it
1. Easy to learn and use	Same layout on every screen; booked slots dimmed (S1)
2. Enhance user productivity	Default slot lengths and office hours; search and group filter (S5, S6)
3. Provide flexibility	In person or Teams; browse any week; filter by meeting type
4. Help and feedback	Help menu; green confirmation; “Last saved” time (S2, S6)
5. Attractive layout and design	Uncluttered cards, one accent colour, colour legend
6. Consistent terminology	Book, Reschedule, Cancel everywhere; red = cancel, green = done
7. Enhance the interface	Calendar control for dates; option buttons for type and location
8. Focus on data entry screens	Form filling, a caption on every field, Back without saving (S2)
9. Use validation rules	Existence, range, validity and combination checks (next slide)
10. Manage data effectively	Data entered once; name and email come from the login

Validation rules stop bad bookings before they are saved

Existence check

Topic and group name are required (marked *)

Range check

Topic is limited to 200 characters, with a live counter

Validity check

Type must be Individual or Group; location In person or Teams

Combination check

A group meeting needs a group name; changes only up to 2 hours before

Data type check

Dates and times are picked from the calendar, never typed

Confirm before deleting

Cancel asks “Yes, cancel” or “Keep it” first (S3)

Garbage in, garbage out: checking input at the source keeps reports and reminders accurate.

All nine Must-have requirements appear in the prototype

Students

See open slots (FR-01) · Book (FR-03) · No double-booking (FR-04) · Confirmation (FR-05) · Reminder (FR-06) · Cancel or reschedule (FR-07)

Instructor

Set availability (FR-02) · Dashboard (FR-08) · Private notes (FR-09)

Also shown

Should-haves: meeting status (FR-10), Teams link (FR-11), Outlook sync (FR-12). Could-have: group history (FR-13)

Requirement IDs match Deliverable 05 – Final Requirements.

RECOMMENDATION

Build it on Microsoft Bookings and go live for Winter 2027

$2,880
one-time cost, then $900 a year

~8 months
payback period

Jan 11
go-live, after testing with Walter

Next: Walter reviews the prototype → we revise → IT Services enables Bookings → class demo and quick reference guide